

# Text Classification in NLP

CMPE346 · Group 2

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## Group task

Text Classification

## Metric

Accuracy & Weighted F1

| Student     | Task                 |
|-------------|----------------------|
| Atakan Gül  | Media Bias Detection |
| [Student 2] | [Topic]              |
| [Student 3] | [Topic]              |

## BiLSTM

Bidirectional LSTM trained from scratch. Captures sequential context in both directions.

## BERT `bert-base-uncased`

Transformer encoder pretrained via masked language modeling. Fine-tuned end-to-end.

## RoBERTa `roberta-base`

Optimized BERT: removes next sentence prediction, uses dynamic masking and more data.

## Task

Classify a news sentence as *biased* or *not biased*.

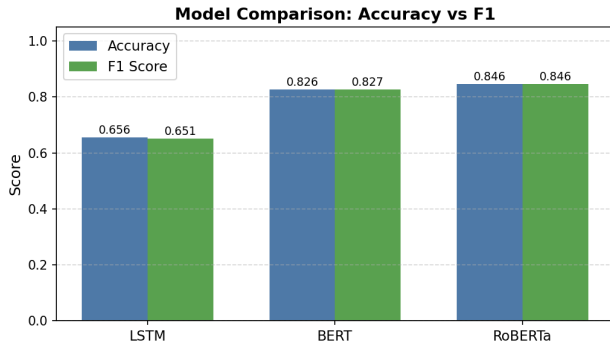
## Dataset: BABE

- 4,121 expert-annotated sentences
- Diverse news outlets
- 3,121 train / 1,000 test

# Atakan — Results

| Model   | Acc.         | F1           |
|---------|--------------|--------------|
| BiLSTM  | 0.656        | 0.651        |
| BERT    | 0.826        | 0.827        |
| RoBERTa | <b>0.846</b> | <b>0.846</b> |

- RoBERTa best: +19% over BiLSTM
- BERT close behind
- BiLSTM struggles on subtle bias



# Atakan — Prediction Examples

| Sentence                                                                                      | True | LSTM | BERT | RoBERTa |
|-----------------------------------------------------------------------------------------------|------|------|------|---------|
| <i>“Nearly 78% of Americans report concern about the climate...”</i>                          | NB   | NB ✓ | NB ✓ | NB ✓    |
| <i>“Companies like Nike and Google made June-teenth a paid holiday.”</i>                      | NB   | B ×  | NB ✓ | NB ✓    |
| <i>“None of Biden’s Democrat counterparts challenged his claim of economic selflessness.”</i> | B    | B ✓  | NB × | B ✓     |

NB = Not Biased    B = Biased    ✓ = correct    × = incorrect

- Media bias is expressed through framing and tone — not facts
- Pretraining is critical: transformers capture patterns BiLSTM cannot
- RoBERTa's optimized pretraining gives it a clear edge over BERT

| Task          | Dataset: [Name]    |
|---------------|--------------------|
| [Description] | Stats              |
|               | Train / test split |

## [Student 2] — Results

| Model   | Acc. | F1 |
|---------|------|----|
| BiLSTM  | —    | —  |
| BERT    | —    | —  |
| RoBERTa | —    | —  |

*[Figure]*

ling 1

ling 2

ding 1

ding 2

ding 3

| Task          | Dataset: [Name]    |
|---------------|--------------------|
| [Description] | Stats              |
|               | Train / test split |

## [Student 3] — Results

| Model   | Acc. | F1 |
|---------|------|----|
| BiLSTM  | —    | —  |
| BERT    | —    | —  |
| RoBERTa | —    | —  |

*[Figure]*

ling 1

ling 2

ding 1

ding 2

ding 3

# All Results

| Task        | Model   | Acc.         | F1           |
|-------------|---------|--------------|--------------|
| Media Bias  | BiLSTM  | 0.656        | 0.651        |
|             | BERT    | 0.826        | 0.827        |
|             | RoBERTa | <b>0.846</b> | <b>0.846</b> |
| [Student 2] | BiLSTM  | —            | —            |
|             | BERT    | —            | —            |
|             | RoBERTa | —            | —            |
| [Student 3] | BiLSTM  | —            | —            |
|             | BERT    | —            | —            |
|             | RoBERTa | —            | —            |

- RoBERTa consistently outperforms across all three tasks
- BERT and RoBERTa adapt well regardless of classification domain
- Pretraining is decisive — BiLSTM cannot compete on small datasets